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Complexity Adjusted Comparisons of LPs and VARs: Evidence from Monte-Carlo Simulations

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1. Introduction (1.5-2 pages)

To infer statements about the dynamic response to exogeneous shocks, the estimation of the corresponding impulse response has relied on two fundamental approaches, either through vector autoregressive (VAR) or Local Projection (LP) models. Since the introduction of the latter, an extensive literature has emerged discussing the relative merits of the two. A first strand of literature has established that, under relatively general conditions, LPs and VARs estimate the same population impulse responses (Plagborg-Moller & Wolf, 2021; Fusari et al. 2026). However, in the discussion of finite sample behavior, LPs were often viewed as less sensitive to dynamic misspecification whereas VARs were seen as potentially more statistically efficient (Li et al., 2024; Montiel Olea et al., 2025). Ludwig (2026) offers an analytical approach of explaining the observed finite sample bias-variance tradeoff. He poses that LPs can be represented as a sequence of VAR estimations increasing in lag length as the estimated horizon increases and thus argues for a renewed thinking about how we can compare LPs and VARs – not by lag length, but by effective complexity.

The first part of this paper aims to build on the discussion outlined above by asking whether conventional finite-sample conclusions are robust to alternative benchmark comparisons that relax the standard equal-lag restriction. To answer this question, the paper operationalizes Ludwig's (2026) central implication through a complexity-adjusted benchmarking exercise in which a fixed LP estimator is compared to progressively higher-order VAR specifications that proxy increasing effective dynamic flexibility (De Graeve & Westermark, 2025). This comparison is conducted across empirically relevant sample sizes (see e.g. Herbst & Johannsen 2021) to assess how the resulting bias–variance trade-off depends on the available data.

The central research question is: Do LPs retain their apparent robustness advantage once they are compared not only to equal-lag VARs, but also to higher-order VARs that approximate the effective complexity implied by LP estimation? The contribution is not to establish a new LP–VAR equivalence result, but to ask whether standard Monte Carlo comparisons between LPs and equal-lag VARs are affected by the fact that LPs correspond to a more flexible dynamic benchmark.

The second part of the paper introduces controlled dynamic misspecification. While the baseline experiment studies complexity-adjusted LP–VAR comparisons under correctly specified linear dynamics, the main practical argument for LPs concerns their robustness when the estimated linear model is only an approximation to the true DGP (Montiel Olea et al., 2021). To study this channel explicitly, the paper introduces a controlled degree of misspecification into the DGP and examines how the relative performance of LPs, equal-lag VARs, and higher-order VARs chang-es as misspecification becomes more severe.

The simulation design proceeds in three steps. First, we consider a correctly specified bivariate VAR(4) DGP, allowing LP(4), VAR(4), and higher-order VARs to be compared under common autoregressive information. Second, we introduce dynamic misspecification by adding a mov-ing-average shock component, thereby generating VARMA(4,1) dynamics while continuing to estimate finite-order VARs and LPs. Third, we introduce nonlinear misspecification through quadratic terms in the DGP while retaining linear estimators. This structure separates the role of dynamic approximation error from nonlinear functional-form misspecification and allows us to study whether conventional LP–VAR finite-sample rankings are robust to complexity-adjusted VAR benchmarks.

For each estimator, horizon, DGP, and sample size, we compute bias, variance, mean squared error, and empirical coverage rates. Bias and variance describe the finite-sample trade-off in point estimation, while coverage assesses whether the corresponding inference procedures re-main reliable under increasing dynamic or nonlinear misspecification.

2. Theoretical Background (total: 5-6.5 pages)

Impulse response functions can be nonparametically defined as the difference between two conditional forecasts of the outcome variable (e.g. Koop (1996)). Both forecasts are conditioned on the same history up to period $t-1$, but differ in the realization at time t. While one conditions on a shock of size δ to the variable of interest, the other one does not. Following Ramey (2016) shocks hereby refers to a significant exogenous force representing unanticipated movements in exogeneous variables, that is uncorrelated with other current and lagged control variables and other exogeneous shocks. The observed difference in the projected paths at horizon h then identifies the causal effect of the shock in population:

$$\mathrm{IR}(h,\delta) \equiv \mathbb{E}\left[y_{t+h} | s_t = s_0 + \delta; \mathbf{x_t}\right] - \mathbb{E}\left[y_{t+h} | s_t = s_0; \mathbf{x_t}\right] \tag{1}$$

Vector Autoregressions and Local Projections serve as two complementary approaches to estimating impulse responses in finite samples. The former, introduced by Sims (1980), parametrizes the joint dynamics of all variables in a single autoregressive system and impulse responses are then obtained as iterated forecasts from the estimated companion matrix. The latter, established by Jordà (2005), takes a direct forecasting approach by estimating the responses at each horizon independently via OLS, projecting the future outcome directly onto current and lagged values of the data. The two approaches compound estimation error differently across horizons and, at equal lag order, occupy opposite ends of a bias-variance spectrum in finite samples (Li et al., 2024; Olea et al., 2025). Local projections tend toward higher variance, vector autoregressions toward higher bias under misspecification. More recently, however, Ludwig (2026) shows that the equal-order comparison is itself misleading. At a given lag order p, LP(p) is strictly more general than VAR(p), so the conventional bias-variance trade-off conflates the choice of estimator with the choice of model complexity.

2.1 Introduction to VARS (1.5 - 2 pages)

Following Lütkepohl (2005), we start by defining a zero mean VAR(p) population model of the form where the outcome of interest, y_t, is a $K \times 1$ random vector, A_i are fixed $K \times K$ coefficient matrices and u_t are a K dimensional white noise processes with $\mathbb{E}\left[u_t\right] = 0$, $\mathbb{E}\left[u_t {u_t}'\right] = \Sigma_u$, i.e. being a nonsingular covariance matrix, and $\mathbb{E}\left[u_t {u_s}'\right] = 0$ for $t \neq s$. Instead of writing out the full lag structure model the model, we can rewrite the full VAR(p) in its corresponding VAR(1) companion form such that

$$Y_t = \mathbf{A} Y_{t-1} + U_t \tag{2}$$

where

$$Y_t := \begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{bmatrix}_{Kp\times 1}, \mathbf{A} := \begin{bmatrix} A_1 & A_2 & \cdots & A_{p-1} & A_p \\ I_K & 0 & \cdots & 0 & 0 \\ 0 & I_K & & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_K & 0 \end{bmatrix}_{Kp\times Kp}, U_t := \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{Kp\times 1}.$$

Assuming Y_t satisfies the stability condition such that all roots, z, of the characteristic equation lie strictly outside the complex unit circle, the process is covariance stationary and admits a Wold moving-average representation:

$$det(I_{Kp} - \mathbf{A}z) \neq 0 \text{ for } |z| \leq 1 \iff Y_t = \sum_{l=0}^{\infty} \mathbf{A}^l U_{t-l}. \tag{3}$$

Restricting attention to the top K rows of the Wold MA representation yields the moving-average representation for y_t:

$$y_t = \sum_{\ell=0}^{\infty} \Psi_\ell \, u_{t-\ell}, \qquad \Psi_\ell := [\mathbf{A}^\ell]_{1:K,\,1:K} \tag{4}$$

where Ψ_h is the reduced-form impulse response function at horizon h, measuring the response of y_t to a unit reduced-form innovation u_{t-h}. It corresponds to the upper-left $K \times K$ block of $\mathbf{A}^h$, which follows directly from the fact that $U_{t-i} = (u_{t-i}, 0, \ldots, 0)'$ has nonzero entries only in its first K positions. Since the components of u_t are generally contemporaneously correlated, $\Sigma_u = \mathbb{E}(u_t u_t') \neq I_K$, a unit shock to one component of u_t will in general be accompanied by simultaneous movements in the other components. Consequently, the reduced-form impulse responses Ψ_h do not admit a structural interpretation. Structural identification is achieved by imposing restrictions on the impact matrix B in $u_t = B\varepsilon_t$, where $\varepsilon_t \sim (0, I_K)$ are mutually uncorrelated structural shocks, so that $\Sigma_u = BB'$.

The most common approach is recursive Cholesky identification (Ramey, 2016), which restricts B to be lower triangular with positive diagonal entries, uniquely recovering B as the Cholesky factor of Σ_u. The recursive ordering imposes a causal structure: the variable ordered first responds contemporaneously to all shocks, while the variable ordered last is insulated from all contemporaneous feedback. The diagonal entries b_{ii}, $i = 1, \ldots, K$, measure the direct impact of structural shock i on variable i, while the off-diagonal entries b_{ij}, $1 \leq j < i \leq K$, capture the contemporaneous spillover from structural shock j to variable i. Substituting $u_t = B\varepsilon_t$ into (4) yields the structural moving-average representation

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \, \varepsilon_{t-\ell}, \qquad \Theta_\ell := \Psi_\ell B, \qquad B = \begin{pmatrix} b_{11} & 0 & \cdots & 0 \\ b_{21} & b_{22} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ b_{K1} & b_{K2} & \cdots & b_{KK} \end{pmatrix} \tag{5}$$

where the (i, j)-th element of Θ_h measures the response of variable i to a unit innovation in structural shock j at horizon h. The structural impulse response of variable i to shock j is thus

$$\theta_h^{ij} = [\Psi_h B]_{ij} \tag{6}$$

where $[\cdot]_{ij}$ denotes the (i,j)-th element of a matrix.

Estimation of the structural impulse response $\hat{\theta}_h^{ij}$ proceeds in two stages. In the first stage, the VAR coefficient matrices $\hat{A} = (\hat{A}_1, \ldots, \hat{A}_p)$ are estimated equation-by-equation via OLS, which is efficient under the stability condition and yields

$$\sqrt{T}\left(\mathrm{vec}(\hat{A}) - \mathrm{vec}(A)\right) \xrightarrow{d} \mathcal{N}\big(0,\ \Gamma^{-1} \otimes \Sigma_u\big)\,, \tag{7}$$

where $\Gamma = \mathbb{E}(Z_t Z_t')$ is the second-moment matrix of the stacked regressors $Z_t = (1, y_{t-1}', \ldots, y_{t-p}')'$, and the innovation covariance Σ_u is replaced by its residual analogue $\hat{\Sigma}_u$ (Lütkepohl, 2005).

In the second stage, structural objects are recovered from the reduced-form estimates. The impact matrix $\hat{B}$ is the Cholesky factor of $\hat{\Sigma}_u$, and reduced-form impulse responses are obtained via the recursion

$$\hat{\Psi}_h = \sum_{\ell=1}^{\min(h,p)} \hat{A}_\ell\, \hat{\Psi}_{h-\ell}, \qquad \hat{\Psi}_0 = I_K, \tag{8}$$

so that the structural impulse response estimator $\hat{\theta}_h^{ij} = g(\hat{\alpha}, \hat{\sigma})$, with $\alpha = \mathrm{vec}(A)$ and $\sigma = \mathrm{vech}(\Sigma_u)$, is a smooth nonlinear function of the first-stage estimates,

$$\hat{\theta}_h^{ij} = e_i'\, \hat{\Psi}_h\, \hat{B}\, e_j. \tag{9}$$

Since g is differentiable in $(\hat{\alpha}, \hat{\sigma})$, the delta method yields

$$\sqrt{T}\left(\hat{\theta}_h^{ij} - \theta_h^{ij}\right) \xrightarrow{d} \mathcal{N}\big(0,\ \nabla g(\alpha,\sigma)'\, \Sigma_{\alpha,\sigma}\, \nabla g(\alpha,\sigma)\big)\,, \tag{10}$$

where $\Sigma_{\alpha,\sigma}$ is the joint asymptotic covariance matrix of $(\hat{\alpha}', \hat{\sigma}')'$ and the Jacobian $\nabla g(\alpha, \sigma)$ admits a closed-form expression (Lütkepohl, 1990). The resulting confidence intervals are valid pointwise in h under stationarity, but not uniformly over the persistence of the process nor over growing horizons, motivating a direct examination of finite-sample behaviour.

The favourable asymptotic properties established above are qualified by three interrelated problems that arise in the sample sizes typical of macroeconomic applications. that arise in the sample sizes typical of macroeconomic applications. First, estimation error propagates through iteration. Because $\hat{\Psi}_h$ is the upper-left block of $\hat{\mathbf{A}}^h$, error in $\hat{A}$ enters the impulse response through powers of the companion matrix and is compounded multiplicatively as the horizon grows. Even small coefficient biases therefore translate into substantial distortions of the estimated responses at longer horizons (Kilian, 1998). In the limiting case of unit or near-unit roots, Phillips (1998) shows that long-horizon impulse responses from unrestricted VARs are not even consistently estimated, converging instead to random variables rather than the true responses. Second, the autoregressive coefficients themselves are biased downward in short samples. It is a long-established result that least squares underestimates the persistence of an autoregressive process (Marriott & Pope, 1954; Nicholls & Pope, 1988; Pope, 1990), and because $\hat{\Psi}_h$ inherits this bias through the recursion, the estimated impulse responses decay too quickly and understate the true persistence of the system. The distortion is most severe precisely when persistence is high and the sample is short. Third, these distortions undermine delta-method inference. When the largest roots approach the unit circle, the asymptotic normal approximation deteriorates and the resulting confidence intervals undercover, increasingly so at longer horizons (Kilian, 1998; Kilian, 1999).

Furthermore, a distinct class of distortions arises when the autoregressive structure is itself misspecified: if the true process has moving-average components, a finite-order VAR is generically inconsistent regardless of sample size, and while higher-order AR approximations can asymptotically absorb the omitted MA dynamics, the required lag expansion rapidly exhausts degrees of freedom in samples of typical macroeconomic length (Braun & Mittnik, 1993). The consequences of lag-order choice are strongly asymmetric—overfitting merely inflates variance, whereas underfitting eliminates higher-order dynamics of the response curve and produces spuriously tight intervals with coverage far below nominal levels (Kilian, 2001). Crucially, both channels are population-level phenomena: Ravenna (2007) shows that truncation bias in the VAR coefficients and the resulting contamination of the structural rotation matrix persist even as $T \to \infty$.

2.2 Introduction to Local Projections (1.5 -2 pages)

Whereas the VAR recovers the impulse response by iterating a single fitted model forward through powers of the companion matrix, the local projection estimates each horizon's response directly from a separate regression (Jordà, 2005). Following Jordà (2005) and Jordà and Taylor (2025), going back to our definition of an impulse response in equation (1), consider a structural shock of interest $\varepsilon_{j,t}$ and fix a horizon h. The LP estimates the response by projecting the future outcome y_{t+h} onto the current shock $\varepsilon_{j,t}$, controlling for the p lags $y_{t-1}, \ldots, y_{t-p}$ that span the conditioning information of the system,

$$y_{t+h} = \nu_h + \beta_h \, \varepsilon_{j,t} + \sum_{\ell=1}^{p} \Xi_{h,\ell} \, y_{t-\ell} + \xi_{t+h}^{(h)}, \qquad h = 0, 1, \ldots, H, \tag{11}$$

where ν_h is a $K \times 1$ vector of constants, β_h is a $K \times 1$ vector of impulse response coefficients at horizon h, $\Xi_{h,\ell}$ are $K \times K$ coefficient matrices on the lagged controls, and $\xi_{t+h}^{(h)}$ is the horizon-h projection residual. Under appropriate identification of the shock, e.g. exogeineity, the impulse response to a one-unit shock to the endogeneous variable can be directly obtained by ordinary least squares estimation of Equation 11. The LP thus recovers the impulse response without parametrizing the underlying propagation dynamics, in sharp contrast to the VAR's reliance on a fitted autoregressive model and forward iteration. If the exogeneity is not appropriate causal interpretation of the shock might be enabled through selection on observable covariates (Olea et al., 2025), instrumental variable regression using a slighly adjusted exogeneity assumtion (Jordà et al., 2015; Stock & Watson, 2018) or even traditional VAR identification schemes like the aforementioned Cholesky decomposition (Plagborg-Møller & Wolf, 2021).

Unlike the VAR, estimation of the structural impulse response $\hat{\beta}_h$ proceeds in a single stage, directly by OLS on equation (11). Equation (11) therefore is the h-step predictive regression of y_{t+h} on the shock of interest, whose leading coefficient coincides with the horizon-h impulse response (Jordà & Taylor, 2025). Since $\hat{\beta}_h$ is the OLS coefficient on a regressor orthogonal to the projection error $\xi_{t+h}^{(h)}$ by construction of the structural shock, it recovers the impulse response directly, without propagation through a nonlinear mapping and hence without the delta-method step that governs the VAR estimator in (10).

Provided $\{y_t\}$ is strictly stationary and ergodic with $\mathbb{E}\|y_t\|^r < \infty$ for some $r > 4$ and absolutely summable mixing coefficients, the least-squares estimator is $\sqrt{n}$-asymptotically normal,

$$\sqrt{n}\left(\hat{\beta}_h - \beta_h\right) \xrightarrow{d} \mathcal{N}(0,\ V_h)\,, \tag{12}$$

with consistency $\hat{\beta}_h \xrightarrow{p} \beta_h = \theta_h^{ij}$ following as an immediate corollary (Hansen, 2022). Because the predictive regression aggregates the shocks accruing between t and $t+h$ over overlapping forecast windows, the projection error $\xi_{t+h}^{(h)}$ follows a moving-average process of order $h-1$, thus the regression scores $\hat{\varepsilon}_{j,t}\,\xi_{t+h}^{(h)}$ are therefore serially correlated, and V_h denotes the associated long-run variance.

The long-run form of V_h reflects a structural feature of direct projection. AS VARs confines all dynamic dependence to the one-step innovation and generates longer-horizon responses by iterating a single fitted model, local projections, by contrast, reads each horizon off a separate regression of which the residual absorbs shocks realised between the impulse date and the response date. Inserting the moving-average representation of y_{t+h} into (11) makes this explicit. The projection error collects the structural innovations dated $t+1$ through $t+h$, and is therefore a moving-average process of order $h-1$.

Inference, however, is complicated by the overlapping-horizon structure of the residual: iterating (**??**) shows that $\xi_{t+h}^{(h)}$ is a moving average of the one-step forecast errors accruing between t and $t+h$, and is therefore serially correlated of order $h-1$ even when the innovations are white noise (Jordà, 2005, p. 163). Accordingly Jordà (2005) recommends a heteroskedasticity- and autocorrelation-consistent (HAC) covariance estimator, such as Newey–West (Jordà & Taylor, 2025, p. 75), which delivers pointwise-valid, asymptotically normal inference under stationarity, but—mirroring the delta-method intervals of the VAR—neither uniformly over the persistence of the process nor at horizons that grow with the sample.

Montiel Olea and Plagborg-Møller (2021) show that a minimal modification removes the need for the HAC correction. Augmenting the projection with one additional lag renders the effective regressor of interest stationary even under a unit root; in the leading scalar case the lag-augmented projection

$$y_{t+h} = \beta_h\, y_t + \tilde{\beta}_h\, y_{t-1} + \xi_{t+h}^{(h)} \tag{13}$$

has a coefficient on y_t that equals the coefficient on the residualized innovation u_t in a regression on (u_t, y_{t-1}). The regression score $\xi_{t+h}^{(h)} u_t$ is then serially uncorrelated—because u_t is uncorrelated with its own leads and lags—so that, contrary to conventional wisdom, the ordinary heteroskedasticity-robust Eicker–Huber–White standard error is sufficient despite the serially correlated residual (Montiel Olea & Plagborg-Møller, 2021, pp. 1794–1795). The resulting t-statistic is asymptotically standard normal *uniformly* over $\rho \in [-1, 1]$ and over horizons satisfying $h/T \to 0$, dispensing with the bandwidth and kernel choices of HAC inference and restoring exactly the robustness to persistence and long horizons that VAR inference lacks (Montiel Olea & Plagborg-Møller, 2021, p. 1809); (Jordà & Taylor, 2025, p. 76).

The favourable and robust properties just described are, as for the VAR, qualified in the sample sizes typical of macroeconomic applications, and the qualifications run largely opposite to those of the autoregression.

First, the price of estimating each horizon by an unrestricted regression is variance. The local projection discards the cross-equation and cross-horizon restrictions that the VAR exploits through its recursion, an efficiency loss that is increasing in the horizon (Li et al., 2024). Since the two estimators target the same response in population (Plagborg-Møller & Wolf, 2021, p. 975), no method dominates in mean squared error across data-generating processes; in finite samples the comparison reduces to a bias–variance trade-off in which the VAR is typically more precise at short horizons and the local projection less biased at long ones (Li et al., 2024; Montiel Olea & Plagborg-Møller, 2021).

Second, the local projection is not unbiased in finite samples. Herbst and Johannsen (2024) show that when the data are persistent—the case of interest for impulse-response analysis—LP point estimates can be severely biased at the sample lengths common in the literature, that the bias links the horizons to one another, and that it admits a simple analytical approximation and bias correction. The near-unit-root downward bias familiar from autoregressive estimation thus re-emerges directly in the projection coefficient (Jordà & Taylor, 2025, p. 75).

Third, and offsetting these costs, the single-equation structure confers a robustness the VAR lacks. Because the response is estimated directly rather than by iterating a fitted model, specification errors do not compound across horizons: Jordà et al. (2024) show that in infinite-order processes the local projection has lower bias than the VAR at horizons beyond the truncation lag, precisely because the coefficient is estimated without propagating coefficient error through powers of the companion matrix, which also makes the projection more forgiving of lag-length misspecification (Jordà & Taylor, 2025, p. 65). This robustness is not unqualified on the inference side, however: Herbst and Johannsen (2024) caution that in the same small, persistent samples the autocorrelation-robust standard errors of the classical approach can substantially understate sampling uncertainty—a further argument for the lag-augmented, heteroskedasticity-robust inference of Montiel Olea and Plagborg-Møller (2021).

2.3 On the Equivalence of VARs and Local Projections (2-2.5 pages)

As Local Projection inference gained in popularity over the last couple of years, more and more scholars have started comparing the asymptotical and finite sample trade-offs of both methods. Prior research has established the notion that SVARs are computational more efficient while LPs inherit favorable robustness to misspecification (see e.g.Ramey, 2016; Jordà, 2005). The notion applies in analogy to the comparison of iterated vs. direct forcasting as established by Marcellino et al. (2006).

Plagborg-Møller and Wolf (2021) were the first to actually show that under weak stationarity and unrestriced lag structure both methods estimate the identical impulse response function in population. Therefore any conventional LP impulse response can be obtrained through an appropriately ordered recursive VAR and any VAR impulse response through a LP with appropriate control variables. - they do not impose restrictions of the underlying DGP The logic applies that any VAR model with sufficiently large lag lengths is able to capture the whole covariance properties of the data such that VAR(∞) forecasts coincide with direct LP forecasts. The authors come to the conclusion: - finite lag length, impulse response approximately agree if the same lag length is used for both but differ for $h \geq p$. - in population, linear local projections are exactly as robust to non-linearities in the DGP as linear VARs

Here I will discuss the theoretical results that establish the equivalence of VARs and LPs under certain conditions, as shown by Plagborg-Moller & Wolf (2021) and Fusari et al. (2026). I will also explain how this equivalence is affected by finite sample properties and dynamic misspecification, as discussed by Li et al. (2024), Montiel Olea et al. (2025), and Ludwig (2026).

The conventional finite-sample comparison between local projections and vector autoregressions is typically conducted at equal lag order, pitting LP(p) against VAR(p). Ludwig (2026) shows that this convention rests on a misleading notion of parsimony. His Theorem 1 establishes an exact finite-sample identity: the h-step prediction of an LP(p) is the forward iteration of a sequence of one-step VAR predictions whose lag order rises from p to $p + h - 1$. An equal-order VAR(p) is therefore a restricted special case of LP(p), obtained by holding the lag order fixed along the recursion rather than allowing it to increase. The familiar empirical pattern—lower approximation bias but higher sampling variance for LP relative to a same-order VAR—is, from this vantage point, largely a mechanical consequence of comparing a more general specification with its own restriction, rather than evidence of an intrinsic property of either estimator. This matters because it leaves the standard reading of the bias-variance trade-off, as articulated by Plagborg-Møller and Wolf (2021) and Montiel Olea et al. (2025), confounded: at equal lag order, the method effect (LP versus VAR) cannot be separated from the complexity effect (more versus fewer effective autoregressive parameters).

From Simulations Li et al. (2024): - suggest a stark bias-variance trade-off between LP (low bias, high variance) and moderate-Lag Var estimators (moderatre bias, low variance)

3. Baseline Simulation (4-5 pages)

This confounding motivates the first part of our study. Because Ludwig's contribution is analytical and deliberately abstains from a simulation-based horse race, the question of whether the conventional finite-sample conclusions survive once the equal-lag restriction is relaxed remains open. We therefore operationalize Ludwig's central implication through a complexity-adjusted benchmarking exercise that holds parsimony more nearly fixed, comparing a fixed LP(4) estimator against progressively higher-order VAR specifications that proxy its increasing effective dynamic flexibility. Crucially, because the identity in Theorem 1 maps LP(p) to a sequence of VARs rather than to any single higher-order VAR, no individual VAR is the appropriate counterpart; instead, LP(4) is read as a reference line against the full VAR order frontier, against which its bias, variance, and coverage can be located. We frame the exercise as a test of robustness rather than as a foregone conclusion: the aim is to assess whether the conventional bias-variance verdict persists, dissolves, or reverses once the comparison is placed on more comparable terms, and to identify the features of the data-generating process under which each procedure is favored.

3.1 Simulation Design

AIM: This simulation study examines whether conventional finite-sample conclusions about the bias-variance tradeoff between Local Projections and Vector Autoregressions are robust to relaxing the equal-lag restriction. Specifically, we compare a fixed LP(4) estimator against a sweep of VAR specifications of increasing order, ranging from VAR(1) to VAR(p+H-1), thereby constructing a complexity-adjusted benchmark in the spirit of Ludwig (2026). The central question is whether LP(4) retains its finite-sample MSE advantage over the equal-lag VAR(4) once the VAR is allowed to exploit its natural complexity advantage at longer horizons.

DGP: The baseline DGP is a stable bivariate zero-mean VAR(4) with recursive Cholesky identification. Recursive identification refers to the lower-triangular structure of the impact matrix B, obtained via Cholesky decomposition of the reduced-form error covariance matrix $\Sigma_u = BB'$. This imposes a contemporaneous zero restriction: the first variable does not respond on impact to shocks in the second variable, while the second variable may respond to both structural shocks simultaneously.

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t, \quad u_t \sim (0, \Sigma_u), \quad y_t \in \mathbb{R}^2 \tag{14}$$

$$\Sigma_u = BB', \qquad B = \begin{pmatrix} b_{11} & 0 \\ b_{21} & b_{22} \end{pmatrix}, \quad b_{11}, b_{22} > 0 \tag{15}$$

$$\rho(\mathcal{A}) < 1, \qquad \mathcal{A} = \begin{pmatrix} A_1 & A_2 & A_3 & A_4 \\ I_2 & 0 & 0 & 0 \\ 0 & I_2 & 0 & 0 \\ 0 & 0 & I_2 & 0 \end{pmatrix} \in \mathbb{R}^{8\times 8} \tag{16}$$

Persistence is calibrated via the spectral radius of the companion matrix across the empirically relevant range $\rho \in (0.5, 0.95)$. The lower bound reflects the observation that, for weakly persistent processes, impulse responses decay rapidly toward zero at longer horizons, rendering the bias-variance comparison between LP and VAR uninformative. The upper bound approaches the unit-root boundary while maintaining covariance stationarity, covering the high-persistence region where finite-sample differences between estimators are most pronounced.

$$A_1 \leftarrow \frac{\rho}{\rho(\mathcal{A})} \cdot A_1, \quad \rho \in (0.5,\, 0.95) \tag{17}$$

Sample sizes are set to $T \in \{100, 200, 240, 500\}$, covering the range of quarterly macroeconomic samples typically encountered in applied work. The choice follows Herbst and Johannsen (2021), who document that the modal sample size in the structural VAR literature lies between 100 and 240 quarterly observations. The value $T = 500$ serves as a large-sample reference point to assess convergence toward the population equivalence result of Plagborg-Møller and Wolf (2021).

The structural estimand is $\theta_h = \Psi_h B e_1$, where Ψ_h denotes the h-step ahead moving-average matrix of the reduced-form VAR, defined via $y_{t+h} = \sum_{j=0}^{\infty} \Psi_j u_{t+h-j}$ with $\Psi_0 = I_2$, B is the Cholesky impact matrix, and $e_1 = (1, 0)'$ is a selection vector. The primitive shock is normalized as a unit shock: we trace the dynamic response of the system to a one-standard-deviation increase in the first structural innovation ε_{1t}. Since B is normalized such that $\mathrm{Var}(\varepsilon_t) = I_2$, no further rescaling is required.

$$u_t = B\varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2) \tag{18}$$

The innovation term is specified as Gaussian white noise. The reasoning behind Gaussianity is two-fold. First, our primary performance measures are bias, variance, and RMSE of the IRF estimates, which in population depend exclusively on the second moments of the underlying process, i.e. the VAR coefficients and the covariance matrix Σ_u. The distribution of the innovation term is therefore irrelevant for the point estimator comparison. Second, we follow Plagborg-Møller and Wolf (2021), who introduce Gaussianity explicitly as an expositional device that allows conditional expectations to be written in place of linear projections. As all results hold under covariance stationarity alone, the distributional assumption leaves the analysis qualitatively unchanged. Importantly, this choice is orthogonal to the misspecification parameters introduced in the extensions: varying the MA parameter θ in Extension 1 or the nonlinearity parameter γ in Extension 2 alters the structure of the DGP while leaving the innovation distribution fixed, ensuring that differences in estimator performance across designs are attributable solely to the respective misspecification channel.

ESTIMAND:

$$\theta_h = \Psi_h B e_1, \qquad e_1 = (1,0)', \qquad h = 1, \ldots, 20 \tag{19}$$

METHODS: We compare two classes of estimators. The first is a fixed LP(4) estimator, which serves as the reference point. At each horizon $h = 1, \ldots, H$, the LP(4) estimator is obtained from the OLS regression

$$y_{1,t+h} = \mu_h + \beta_h x_t + \sum_{\ell=1}^{4} \delta'_{h,\ell} y_{t-\ell} + \xi_{h,t}, \quad h = 1, \ldots, H \tag{20}$$

where $x_t = e_1'(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$ is the estimated structural shock recovered from the first-stage VAR(4). The structural impulse response estimate at horizon h is $\hat{\theta}_h^{\mathrm{LP}} = \hat{\beta}_h$.

The second class consists of a sweep of VAR estimators of increasing order, VAR(q) for $q = 1, \ldots, p + H - 1$, where $p = 4$ and $H = 20$. For each order q, the VAR is estimated equation-by-equation via OLS, the reduced-form covariance matrix $\hat{\Sigma}_u^{(q)}$ is Cholesky-decomposed to recover the structural impact matrix $\hat{B}^{(q)}$, and the h-step ahead structural IRF is computed as

$$\hat{\theta}_h^{\mathrm{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1 \tag{21}$$

where $\hat{\Psi}_h^{(q)}$ denotes the h-step ahead moving-average matrix of the estimated VAR(q). This sweep constitutes the complexity-adjusted benchmark motivated by Ludwig (2026): by Theorem 1 of that paper, LP(4) at horizon h is algebraically equivalent to a sequence of VAR predictions stepping through orders $4, 5, \ldots, 4+h-1$. The fair comparison is therefore not LP(4) against VAR(4) alone, but against the full range of VAR specifications up to order $p+H-1$. Note that this equivalence holds exactly for the sequence of VARs of increasing order; comparing LP(4) to any single VAR(q^*) is an approximation, with the complexity-adjusted single-order approximation given by $q^* = p + \frac{h-1}{2}$ (Ludwig, 2026). The equal-lag VAR(4) is nested in the sweep as a special case and is retained as an additional reference point to connect to the conventional benchmark in the literature.

Identification of the structural shock is implemented as follows. For each VAR(q) in the sweep, the structural impact matrix $\hat{B}^{(q)}$ is obtained via Cholesky decomposition of the order-specific residual covariance matrix $\hat{\Sigma}_u^{(q)}$. For the LP(4), which does not estimate a complete system, identification is achieved through a first-stage VAR(4): the structural shock is constructed as $\hat{\varepsilon}_{1t} = e_1'(\hat{B}^{(4)})^{-1}\hat{u}_t^{(4)}$, consistent with the LP lag order $p = 4$. Since Plagborg-Møller and Wolf (2021) establish that LP and VAR share the same implicit shock in population under recursive identification, this ensures that any divergence in estimated IRFs reflects differences in propagation rather than identification. No data-dependent lag selection is performed within the sweep; all VAR orders $q = 1, \ldots, p + H - 1$ are estimated on each simulated sample, ensuring that the comparison is across fixed specifications rather than across selection procedures.

PERFORMANCE: Performance is assessed along two dimensions, reported horizon-by-horizon for $h = 1, \ldots, 20$. For point estimation, we report bias, variance, MSE, and RMSE of the IRF estimates $\hat{\theta}_h$ relative to the true estimand θ_h, computed across $B = 5{,}000$ Monte Carlo replications. The number of replications exceeds the $B = 1{,}000$ used by Li et al. (2024) and Olea et al. (2025), ensuring tighter Monte Carlo standard errors at the cost of additional computation. Formally:

$$\widehat{\text{Bias}}_h = \frac{1}{B} \sum_{b=1}^{B} \left(\hat{\theta}_h^{(b)} - \theta_h \right) \tag{22}$$

$$\widehat{\text{Var}}_h = \frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\theta}_h^{(b)} - \bar{\hat{\theta}}_h \right)^2 \tag{23}$$

$$\widehat{\text{MSE}}_h = \frac{1}{B} \sum_{b=1}^{B} \left(\hat{\theta}_h^{(b)} - \theta_h \right)^2 \tag{24}$$

$$\widehat{\text{RMSE}}_h = \sqrt{\widehat{\text{MSE}}_h} \tag{25}$$

Monte Carlo standard errors are reported alongside all point-estimator performance measures to quantify simulation uncertainty, with $\text{MCSE}(\widehat{\text{Bias}}_h) = \sqrt{\widehat{\text{Var}}_h / B}$.

For inference, we assess coverage of nominal 95% confidence intervals. For the VAR(q) sweep, confidence intervals are constructed via the delta method, propagating estimation uncertainty in the reduced-form coefficients $(\hat{A}_1, \ldots, \hat{A}_q)$ through the nonlinear mapping to structural IRFs $\hat{\theta}_h^{\text{VAR}(q)} = \hat{\Psi}_h^{(q)} \hat{B}^{(q)} e_1$ (Kilian & Lütkepohl, 2017). For LP(4), confidence intervals are based on heteroskedasticity-robust standard errors, which suffice under the martingale difference assumption on the structural shock (Plagborg-Møller & Wolf, 2021). Coverage is defined as the fraction of replications in which the nominal 95% interval contains the true estimand θ_h:

$$\widehat{\text{Coverage}}_h = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1} \left[\theta_h \in \hat{CI}_h^{(b)} \right] \tag{26}$$

3.2 Simulation Results

Figure 1: TRUE MC Simulated DGPS

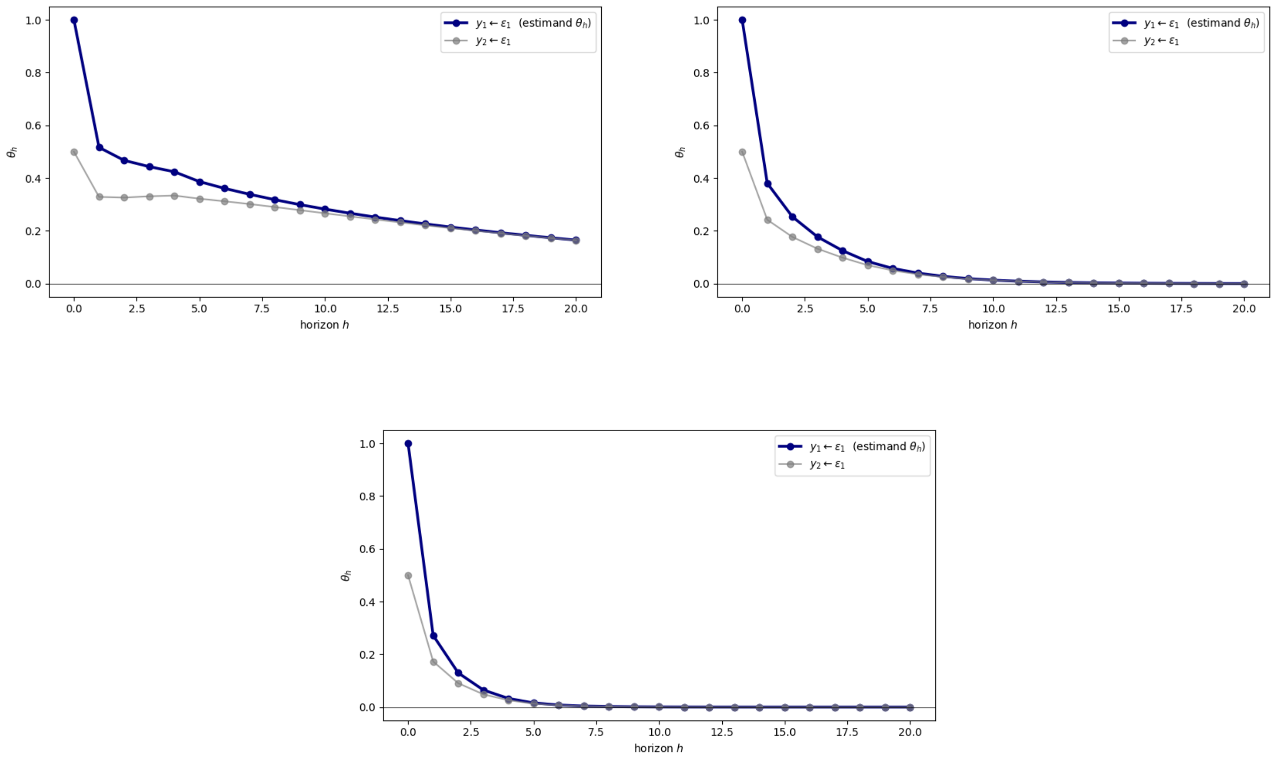


Figure 2: POINT ESTIMATES ACROSS DGPs

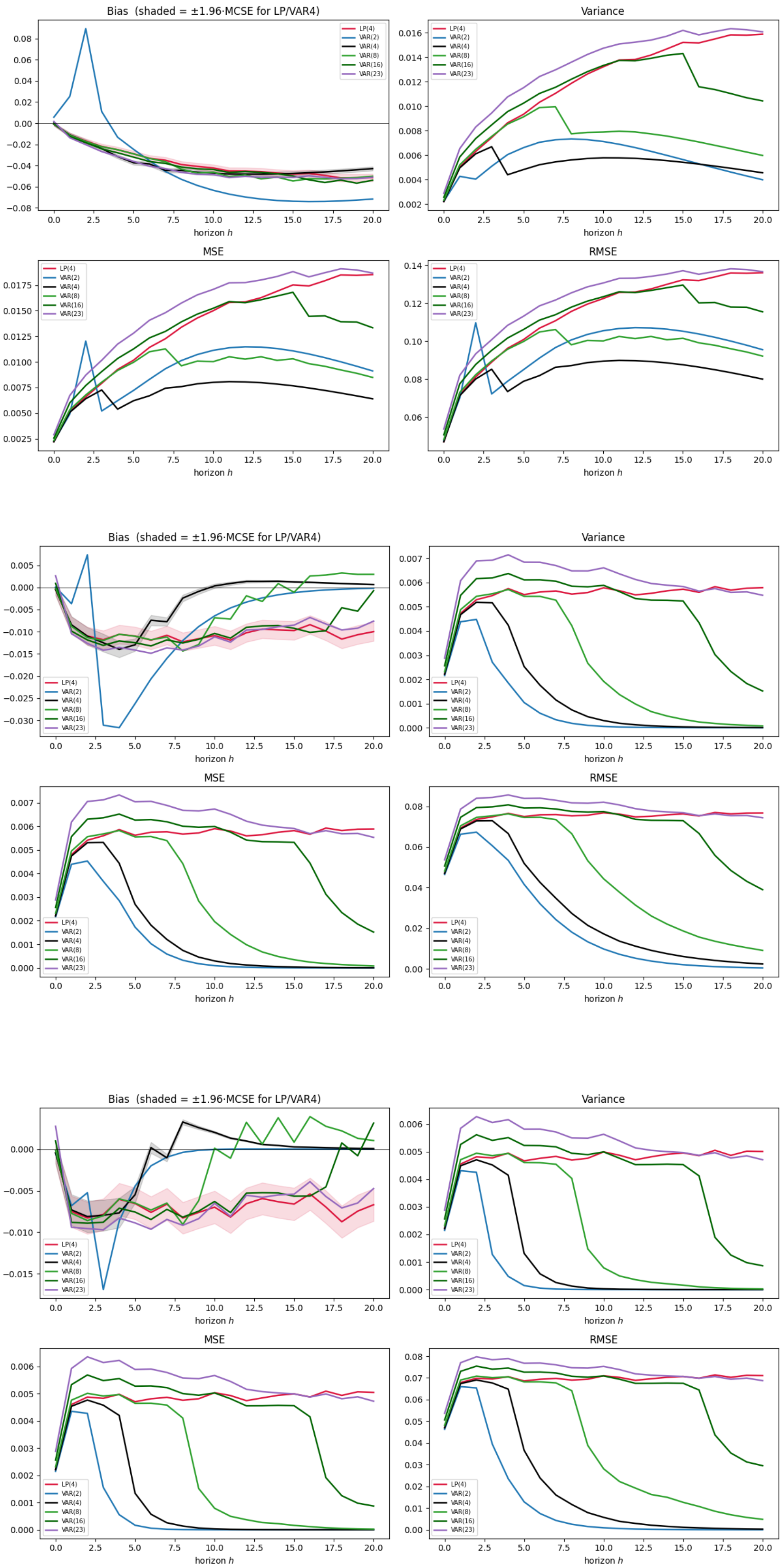
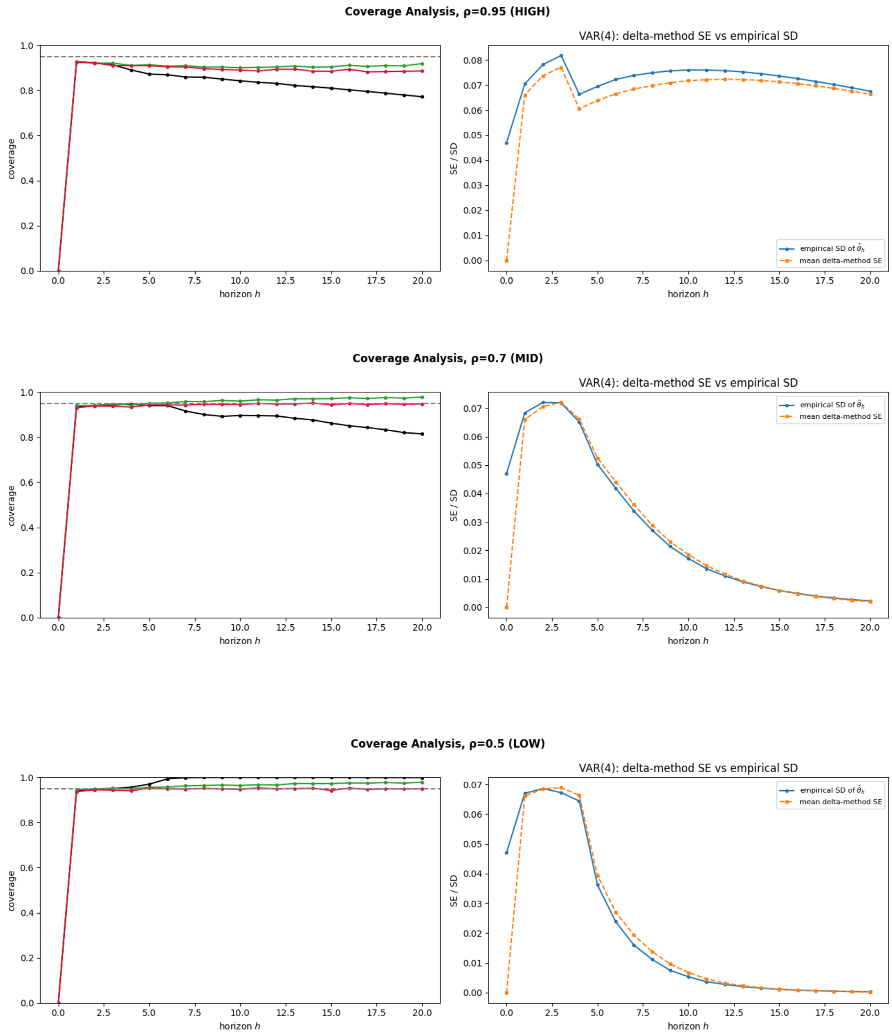


Figure 3: COVERAGE ACROSS DGPs



4. Extension 1: Dynamic Misspecification (3-4 pages)

4.1 Simulation Design

Extension 1 examines whether the finite-sample MSE ranking between LP(4) and the complexity-adjusted VAR sweep is robust to dynamic misspecification of the Wold representation. The baseline VAR(4) is augmented by a first-order moving average component, yielding a VARMA(4,1) DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + u_t + \Theta u_{t-1}, \quad u_t = B\varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2) \tag{27}$$

where $\Theta = \theta I_2$ is a scalar-scaled MA impact matrix, with misspecification strength $\theta \in \{0.1, 0.3, 0.6\}$. The case $\theta = 0$ nests back to the baseline VAR(4), ensuring continuity across designs. The VAR coefficient matrices $A_1, \ldots, A_4$ and the impact matrix B are inherited from the baseline DGP, so that all variation across Extension 1 scenarios is attributable exclusively to the MA component.

The VARMA(4,1) DGP is invertible for all $\theta \in \{0.1, 0.3, 0.6\}$, admitting an infinite-order VAR representation. This means the complexity-adjusted VAR sweep retains its theoretical motivation: higher-order VAR(q) specifications progressively approximate the true dynamics as $q \to \infty$, though at the cost of increasing parameter proliferation in finite samples. The central question is therefore whether LP(4) — which is robust to this misspecification by virtue of its zero asymptotic bias (Olea et al., 2025) — retains its MSE advantage over the equal-lag VAR(4), and where on the complexity-adjusted VAR frontier it sits relative to the baseline.

Persistence is varied across $\rho \in (0.5, 0.95)$ as in the baseline, since the degree to which the MA component distorts finite-sample VAR approximations depends directly on the persistence of the underlying process: at high ρ, shocks decay slowly and the MA term remains influential at longer horizons, amplifying the VAR's approximation bias. Sample sizes follow the baseline design, $T \in \{100, 200, 240, 500\}$. The misspecification strength $\theta \in \{0.1, 0.3, 0.6\}$ is thus varied jointly with persistence and sample size, yielding a three-way design that isolates the marginal contribution of MA misspecification from the persistence and finite-sample channels already documented in the baseline.

The structural estimand $\theta_h = \Psi_h^{\text{VARMA}} B e_1$ is defined with respect to the true VARMA moving-average matrices Ψ_h^{VARMA}, computed analytically from the VARMA companion representation prior to each simulation run. Both estimators — LP(4) and the VAR(q) sweep — are applied without modification to the VARMA-generated data, so that the misspecification is fully absorbed into the finite-sample performance of each estimator rather than accommodated by the estimation procedure. Performance measures are identical to the baseline.

4.2 Simulation Results

5. Extension 2: Functional Misspecification (3-4 pages)

5.1 Simulation Design

Extension 2 examines whether the finite-sample MSE ranking between LP(4) and the complexity-adjusted VAR sweep is robust to functional misspecification of the DGP. The baseline VAR(4) is augmented by a mean-zero quadratic term in the first lag, yielding a nonlinear DGP:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + A_3 y_{t-3} + A_4 y_{t-4} + \gamma \cdot g(y_{1,t-1}) + u_t, \quad u_t = B\varepsilon_t, \quad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, I_2) \tag{28}$$

where

$$g(y_{1,t-1}) = \left(0,\ \left(\frac{y_{1,t-1}}{\sigma_1}\right)^2 - 1\right)' \tag{29}$$

is a standardised, mean-zero quadratic in the first variable's first lag, entering the second equation only, with $\sigma_1 = \sqrt{\text{Var}(y_{1,t})}$ the stationary standard deviation of $y_{1,t}$ under the linear ($\gamma = 0$) VAR, and $\gamma \in \{0, 0.3, 0.6\}$ controlling the degree of nonlinearity.

Importantly, adding a quadratic term in lagged levels can increase the risk of non-stationary or explosive simulated paths, particularly under high shock persistence. For this reason, this extension restricts the persistence parameters to $\rho \in \{0.5, 0.7\}$. This restriction is intended to mitigate violations of covariance stationarity and to keep the estimand θ_h well-defined over the simulated designs. This specification changes the nature of the misspecification relative to the baseline linear VAR(4). The nonlinear term enters the propagation dynamics through a quadratic function of $y_{1,t-1}$, while the structural shocks continue to enter contemporaneously and linearly through $u_t = B\varepsilon_t$. Thus, the exercise does not isolate nonlinear shock transmission, but studies whether the finite-sample MSE ranking between LP(4) and the complexity- adjusted VAR sweep is robust when the true conditional mean is nonlinear in lagged state variables. Because the nonlinear component is a function of lagged outcomes, increasing the lag length of a linear VAR cannot in general eliminate the functional misspecification and the fitted models remain linear projections of a nonlinear data-generating process.

The key theoretical motivation for this extension follows from Plagborg-Møller and Wolf (2021), who establish that linear LP and linear VAR estimate the same best linear approximation to the true IRF in population, regardless of whether the DGP is linear (Proposition 1, p. 960). Under comparable lag information, both estimators thus provide an equal linear approximation of the non-linear DGP, and any finite-sample divergence reflects differences in the bias-variance tradeoff rather than differential robustness to nonlinearity per se.

This makes Extension 2 complementary to Extension 1, as whereas Extension 1 isolates the bias channel under invertible dynamic misspecification, Extension 2 isolates functional-form misspecification in the conditional mean.

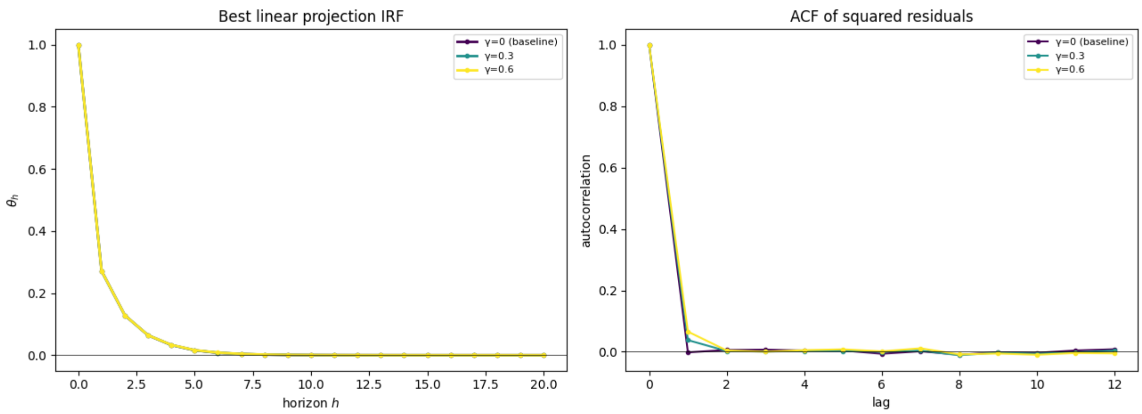
The structural estimand is the best linear approximation to the impulse response generated by the nonlinear DGP, denoted $\theta_h^{\text{lin}} = \Psi_h^{\text{lin}} B^{\text{lin}} e_1$. Here Ψ_h^{lin} denotes the linear projection response implied by the covariance structure of the nonlinear process, and B^{lin} is the corresponding impact matrix obtained from the reduced-form innovation covariance matrix using the same recursive identification scheme as in the baseline design. Since both LP and VAR target the same linear projection estimand in population under the conditions described by Plagborg-Møller and Wolf (2021), the comparison remains well-defined under nonlinear misspecification.

Persistence is varied across $\rho \in \{0.5, 0.7\}$, while sample sizes are chosen to reflect small sample and asymptotic behaviors using $T \in \{240, 10000\}$. The nonlinearity strength $\gamma \in \{0.3, 0.6\}$ is varied jointly with persistence and sample size. Both estimators are applied without modification to data generated by the nonlinear DGP, so the functional-form misspecification is fully reflected in the finite-sample performance of each estimator. Performance measures are identical to the baseline, with added small-sample and asymptotic analysis of deviation from the ground truth IRF, to investigate the claim of equivalence in population.

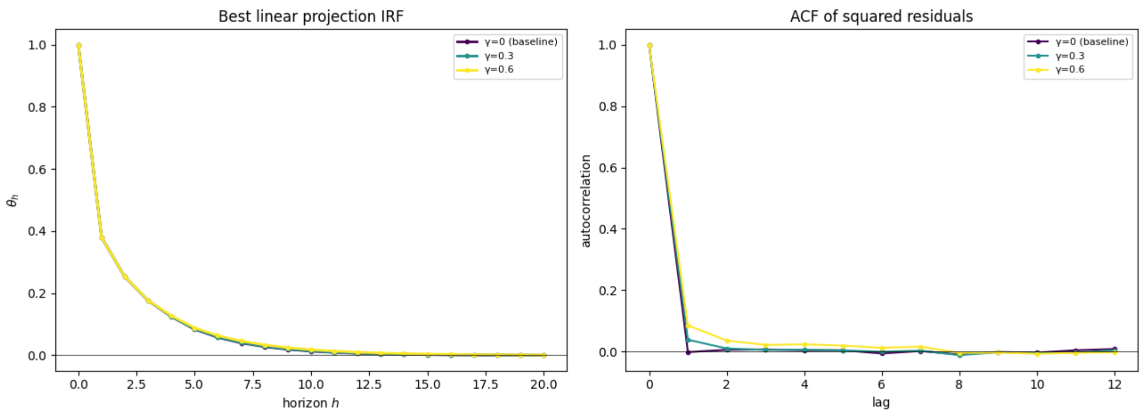
5.2 Simulation Results

Figure 4: True IRF Functions by Nonlinearity Strength

(a) $\rho = 0.5$



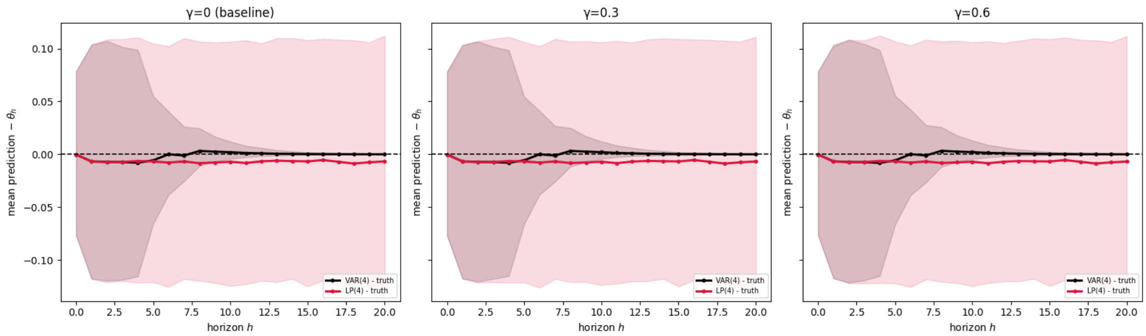
(b) $\rho = 0.7$



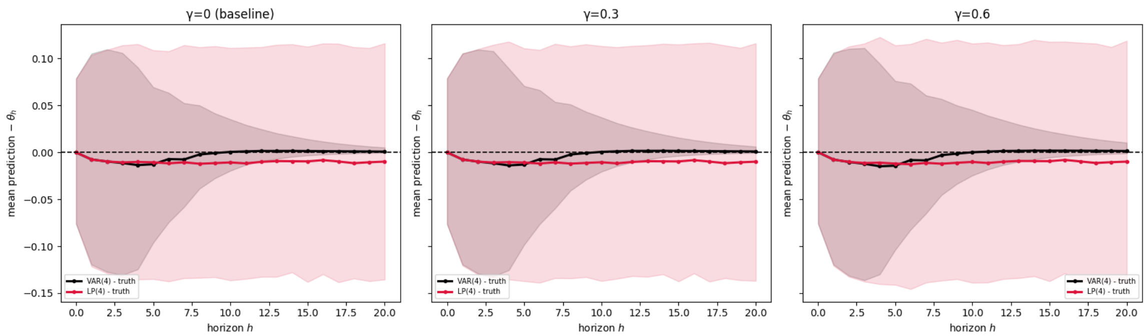
Note. The left column reports the best linear projection impulse response, θ_h, of variable 1 to shock 1. The right column reports the autocorrelation function of the squared second-equation residual, $\hat{e}_{2,t}^2$, as a diagnostic for volatility clustering induced by the nonlinear lag term. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. The best linear projection response is approximately invariant across nonlinearity strengths γ.

Figure 5: Small-Sample Deviation from the True IRF

(a) $\rho = 0.5$



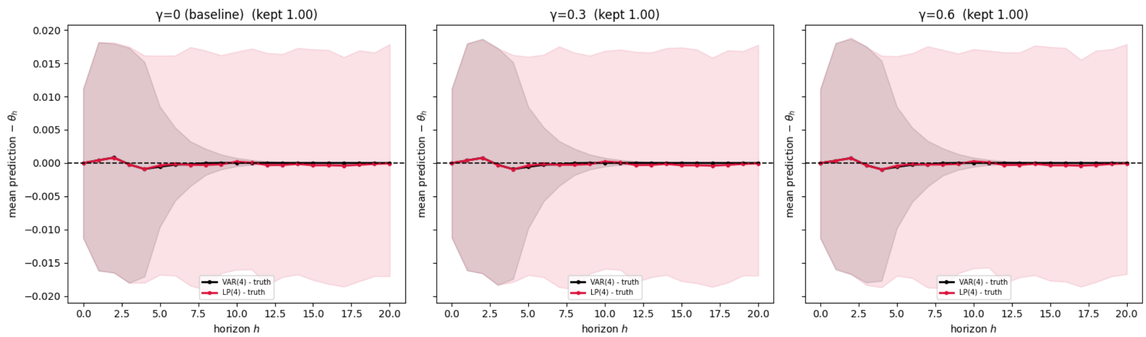
(b) $\rho = 0.7$



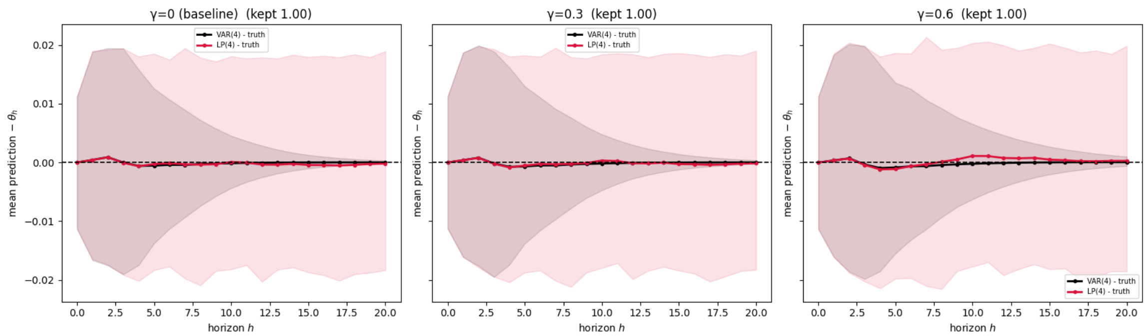
Note. Each panel reports finite-sample deviations from the true impulse response function for the nonlinear lag-quadratic DGP with $T = 240$ and $B = 5{,}000$ Monte Carlo replications. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Columns within each row correspond to nonlinearity strength.

Figure 6: Asymptotic Deviation from the True IRF

(a) $\rho = 0.5$



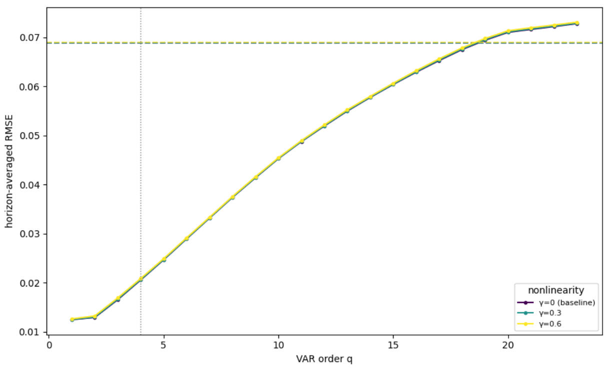
(b) $\rho = 0.7$



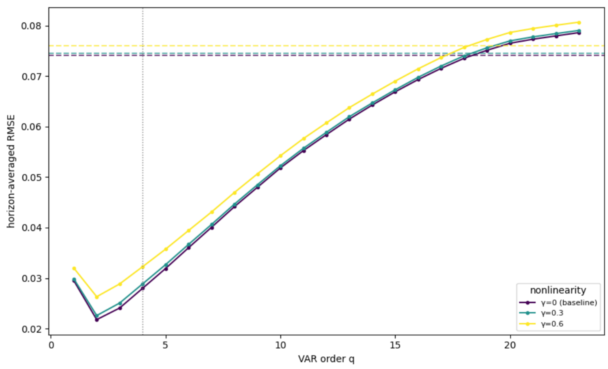
Note. Each panel reports asymptotic deviations from the true impulse response function for the nonlinear lag-quadratic DGP, approximated using $T = 10{,}000$ and $B = 1{,}000$ Monte Carlo replications. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Columns within each row correspond to nonlinearity strength.

Figure 7: Complexity Frontier under Nonlinear Lag Misspecification

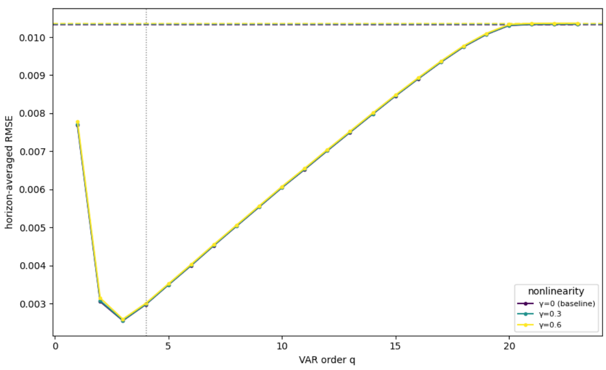
(a) $\rho = 0.5$ (Sample)



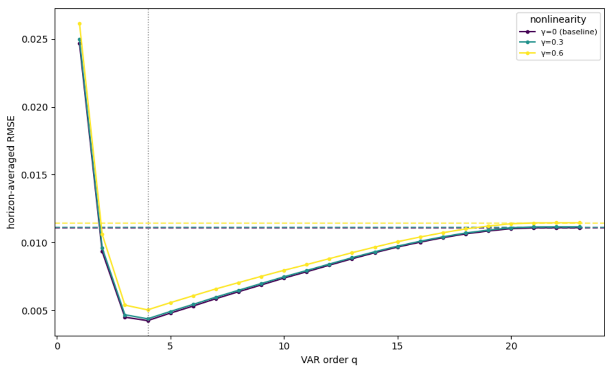
(b) $\rho = 0.7$ (Sample)



(c) $\rho = 0.5$ (Asymptotic)



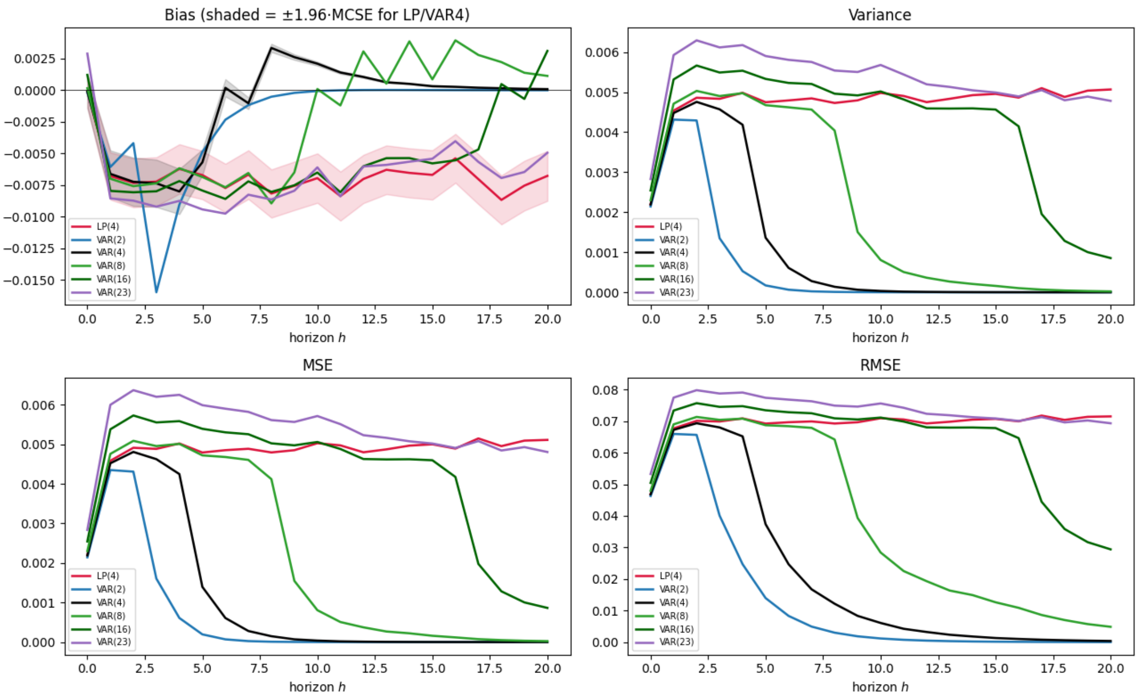
(d) $\rho = 0.7$ (Asymptotic)



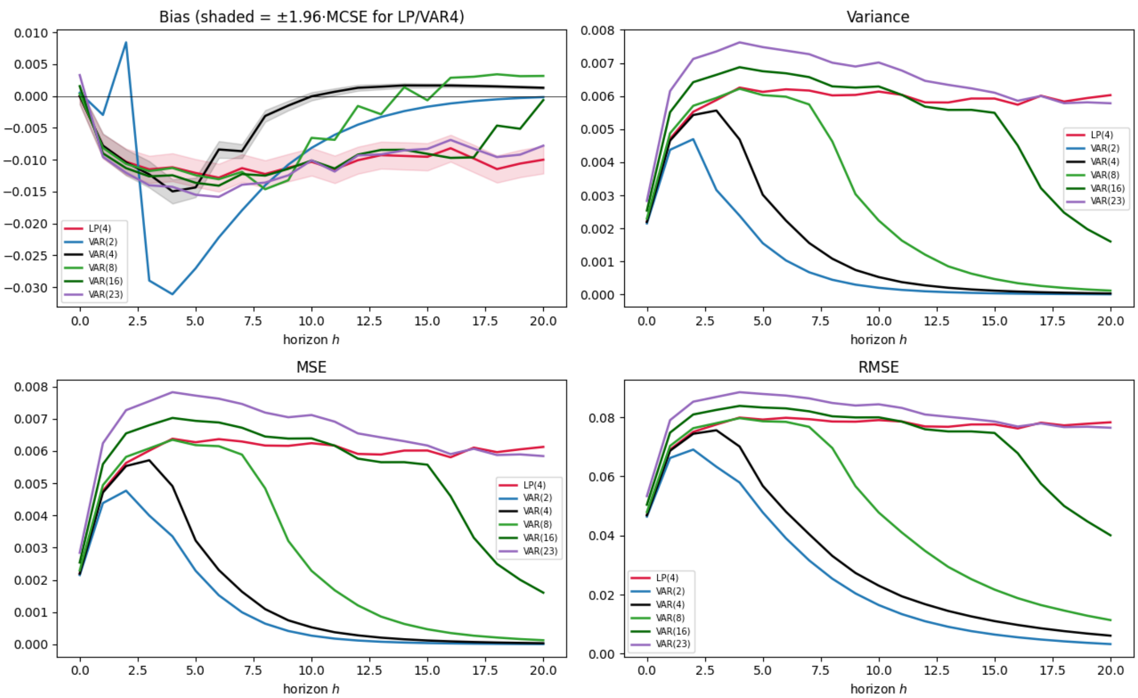
Note. Each panel reports the complexity frontier for the nonlinear lag-quadratic DGP with $T = 240$ for the top row, and $T = 10,000$ for the bottom row. Panels correspond to persistence values $\rho \in \{0.5, 0.7\}$. Columns within each panel correspond to nonlinearity strength. The top row (a), (b), show the small sample complexity frontier, while the bottom row (c), (d), approximates the asymptotic frontier with $T = 10,000$. The frontier summarizes the finite-sample tradeoff between model complexity and estimator performance.

Figure 8: Sample Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$ (Sample)



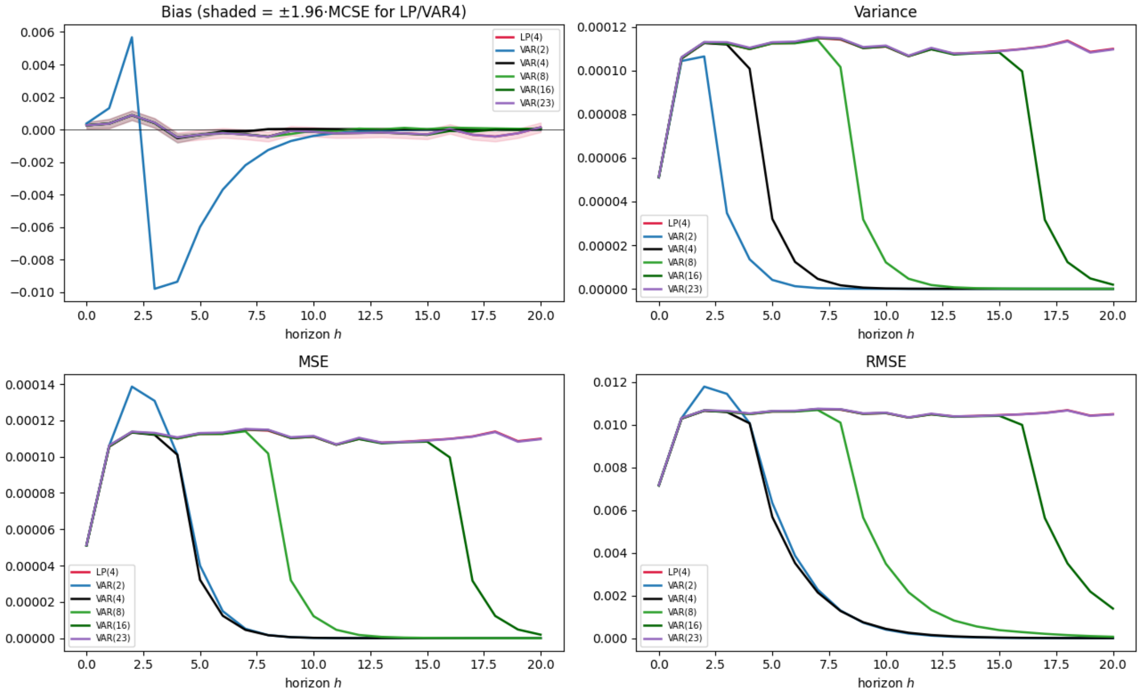
(b) $\rho = 0.7$ (Sample)



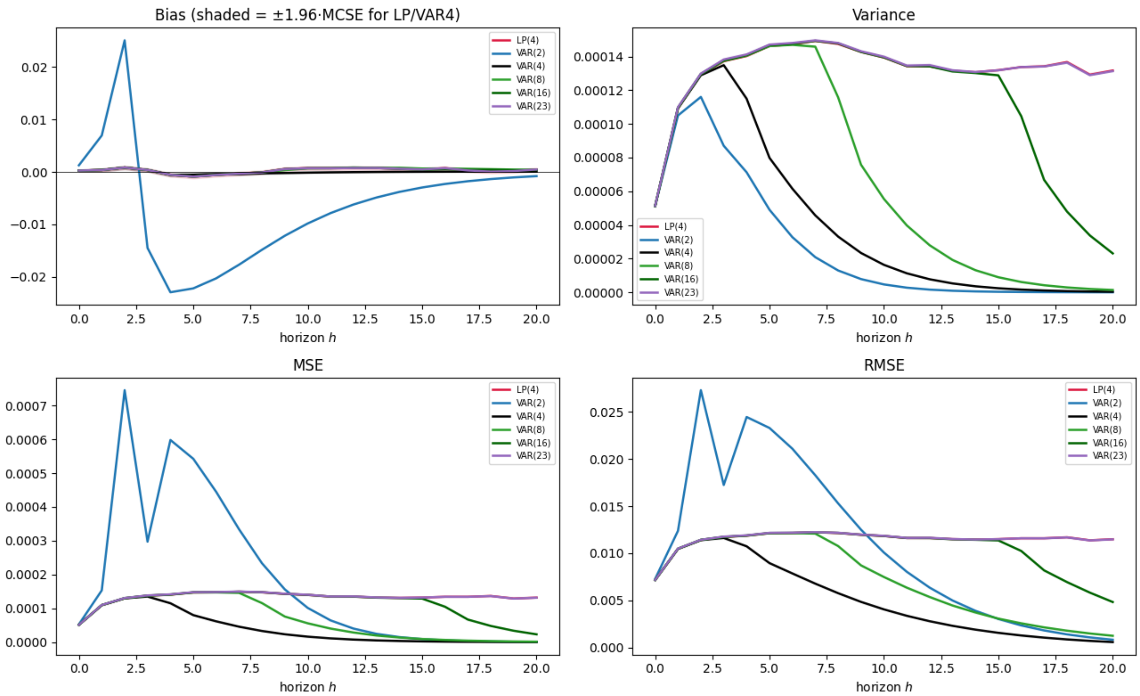
Note. Each panel reports point estimates for the nonlinear lag-quadratic DGP with $T = 240$, to investigate sample behavior of the estimators. $B = 5{,}000$ Monte Carlo replications are performed with $\gamma = 0.6$. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Estimator definitions are identical to the baseline design.

Figure 9: Asymptotic Point Estimates under Strong Nonlinearity

(a) $\rho = 0.5$ (Asymptotic)



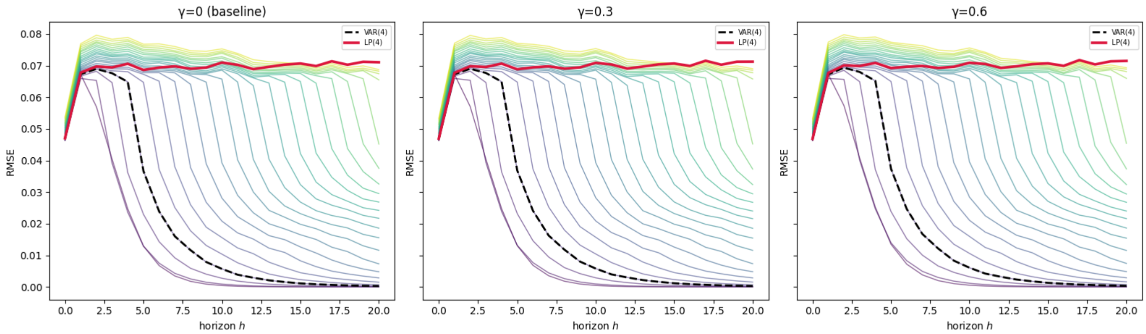
(b) $\rho = 0.7$



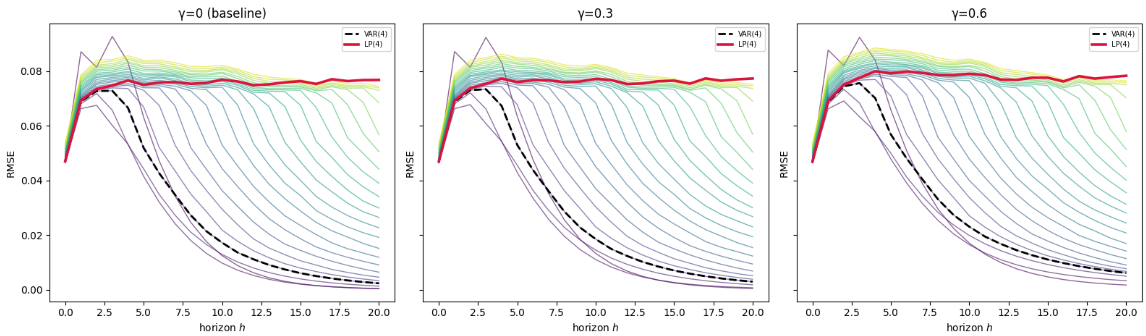
Note. Each panel reports point estimates for the nonlinear lag-quadratic DGP with $T = 10,000$, to approximate asymptotic behavior. $B = 5{,}000$ Monte Carlo replications are performed with $\gamma = 0.6$. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Estimator definitions are identical to the baseline design.

Figure 10: Sample RMSE by Horizon under Nonlinear Lag Misspecification

(a) $\rho = 0.5$



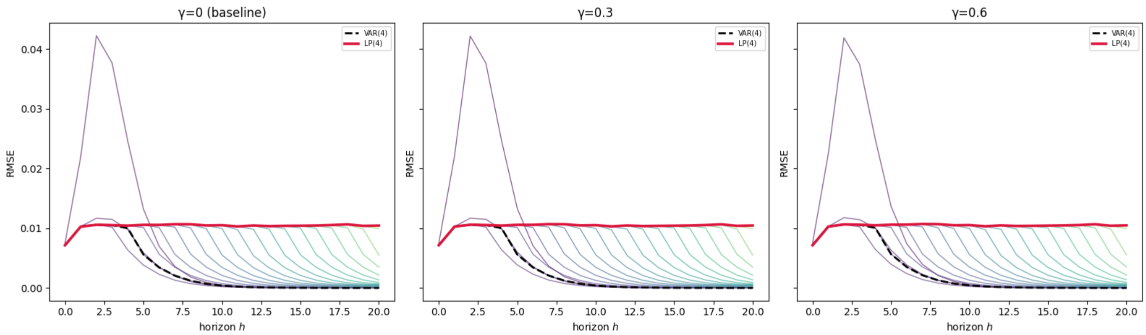
(b) $\rho = 0.7$



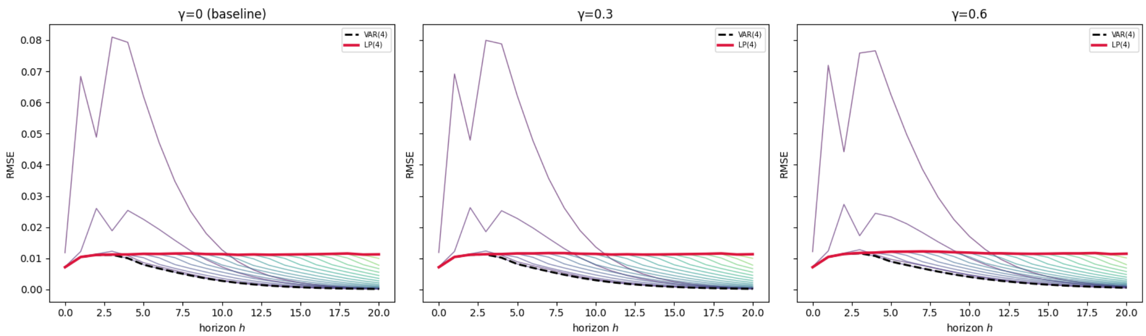
Note. Each panel reports RMSE by horizon for the nonlinear lag-quadratic DGP with $T = 240$. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Columns within each panel correspond to nonlinearity strength γ. Estimator definitions and performance measures are identical to the baseline design.

Figure 11: Asymptotic RMSE by Horizon under Nonlinear Lag Misspecification

(a) $\rho = 0.5$

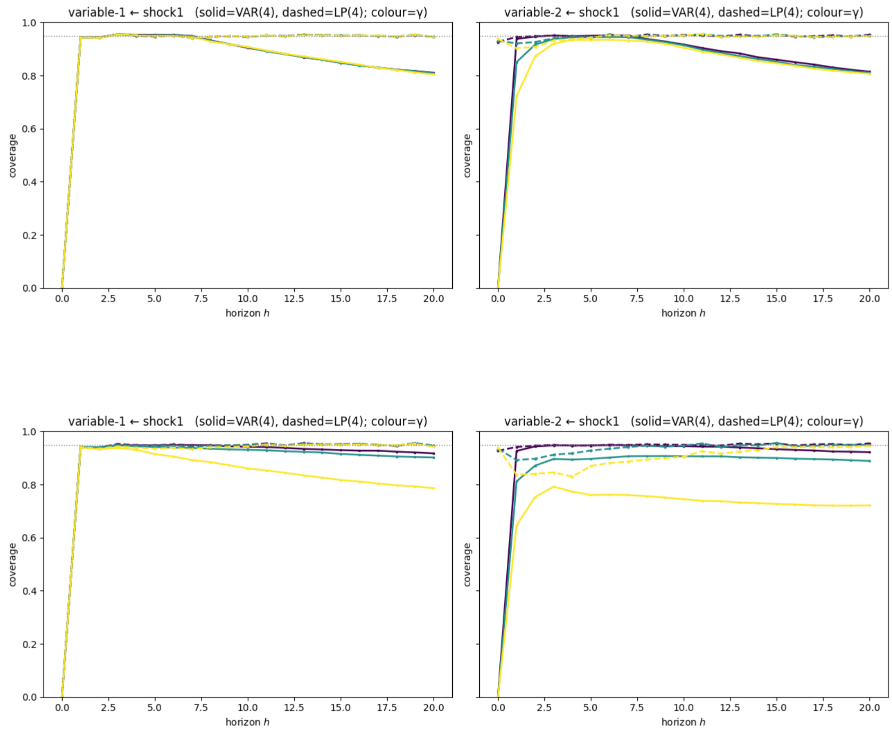


(b) $\rho = 0.7$



Note. Each panel reports RMSE by horizon for the nonlinear lag-quadratic DGP with $T = 10,000$ to approximate asymptotic behavior. Rows correspond to persistence values $\rho \in \{0.5, 0.7\}$. Columns within each panel correspond to nonlinearity strength γ. Estimator definitions and performance measures are identical to the baseline design.

Figure 12: COVERAGE



6. Discussion (1.5-2 pages)

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Declaration of independence